

Attention-Augmented Scale Hyperprior for Learned Image Compression

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1 Introduction

Image compression is an important problem in computer vision because images must often be stored or transmitted under limited bandwidth and storage. Classical codecs such as JPEG use hand-designed transforms and entropy coding, but in recent years learned image compression methods have become strong alternatives. These methods use neural networks to optimize the full compression pipeline end-to-end.

In this project, I want to work on learned image compression by modifying an existing deep model rather than building a completely new one from scratch. I will use the scale hyperprior model of Ballé et al. [1] as my baseline, since it is a standard and well-known model in learned compression. My goal is to add lightweight attention modules to this baseline and test whether they improve rate-distortion performance.

2 Problem Statement

The goal of this project is to improve a standard learned image compression model by adding lightweight attention to its transform network.

Given an input image x , the model compresses it into latent variables and reconstructs an output image $\hat{x}$. The model should produce a reconstruction with high visual quality while using as few bits as possible.

The main question I want to answer is, can lightweight attention improve the rate-distortion performance of the scale hyperprior model?

More specifically, I want to test whether adding a simple attention mechanism helps the encoder and decoder focus more on important image structures such as edges and textures, and whether this leads to better reconstruction quality at the same bitrate.

3 Dataset and Evaluation

For training, I plan to use the DIV2K dataset [3], which is commonly used in low-level vision tasks and contains high-quality images suitable for learned compression.

For evaluation, I plan to use the Kodak dataset [4], which is a standard benchmark in image compression research. This makes the results easier to compare with existing work.

I will evaluate the model using the standard compression metrics:

- bits per pixel (bpp),
- PSNR,
- MS-SSIM.

The main results will be shown as rate–distortion curves:

- bpp vs. PSNR,
- bpp vs. MS-SSIM.

I may also report the number of model parameters and briefly discuss complexity, since attention modules can increase model size slightly.

4 Technical Approach

Baseline Model

My baseline will be the scale hyperprior model proposed by Ballé et al. [1]. This model is a learned image compression architecture that uses an analysis transform (encoder), a synthesis transform (decoder), and a hyperprior branch for entropy modeling.

At a high level, the encoder maps the input image x into a latent representation y . Then a second latent representation z is derived from y and used as side information. This hyperprior helps estimate the probability distribution of the main latent variables more accurately, which improves entropy coding and reduces bitrate.

My Proposed Modification

Instead of changing the entire compression system, I want to make one controlled modification. I will add lightweight attention modules to the transform network.

The idea is simple. Not all image regions are equally important for compression. Smooth areas are easier to represent, while textured regions and edges are harder. A lightweight attention block can help the network emphasize more informative features and suppress less useful ones.

In this project, I want to keep the hyperprior itself unchanged and only modify the encoder/decoder side. I think this is a better design choice for a term project because it makes the comparison cleaner. If the model improves, I can say that the gain came from the attention mechanism.

Planned Model Variants

I plan to compare the following models:

1. the original `bmsbj2018_hyperprior` baseline,

2. a version with attention in the encoder only,
3. a version with attention in both the encoder and decoder.

Implementation Plan

I plan to implement the project in PyTorch using CompressAI [2], which already provides the baseline model and training tools for learned compression. This reduces implementation risk and lets me focus on the actual model modification.

5 Expected Outcome

I do not expect my modification to produce a dramatic breakthrough. A more realistic goal is a small but consistent improvement over the baseline, such as:

- slightly higher PSNR or MS-SSIM at the same bitrate,
- or slightly lower bitrate at similar reconstruction quality.

Even if the attention model does not clearly outperform the baseline, I think the project will still be valuable because it will provide a clean comparison and show whether this type of lightweight attention is useful in this compression setting.

References

- [1] Johannes Ballé, David Minnen, Saurabh Singh, Sung Jin Hwang, and Nick Johnston. Variational Image Compression with a Scale Hyperprior. In *International Conference on Learning Representations (ICLR)*, 2018.
- [2] Jean Bégaint, Fabien Racapé, Simon Feltman, and Akshay Pushparaja. CompressAI: a PyTorch Library and Evaluation Platform for End-to-End Compression Research. *arXiv preprint arXiv:2011.03029*, 2020.
- [3] Radu Timofte, Eirikur Agustsson, Luc Van Gool, Ming-Hsuan Yang, and others. NTIRE 2017 Challenge on Single Image Super-Resolution: Dataset and Study. In *CVPR Workshops*, 2017.
- [4] Kodak Lossless True Color Image Suite. <https://r0k.us/graphics/kodak/>